

STATISTIC WEEK 3

Exercise 3.1

$$a) \sum p(n) = 0.1 + 0.05 + 0.45 + 0.32 = 0.92 \rightarrow \text{Not VALID}$$

$$b) \sum p(n) = 0.35 + 0.28 + 0.23 + 0.14 = 1 \rightarrow \text{VALID}$$

$$b) \sum p(n) = -0.25 + 0.85 + 0.4 = 1 \rightarrow \text{VALID}$$

Exercise 3.2

$$f(n) = \frac{m}{2^n} \quad \text{for } n = 1, 2, 3, 4, 5$$

$$= \frac{m}{2^1} + \frac{m}{2^2} + \frac{m}{2^3} + \frac{m}{2^4} + \frac{m}{2^5} = 1.0$$

$$\frac{m}{31} = \frac{m}{2} + \frac{m}{4} + \frac{m}{8} + \frac{m}{16} + \frac{m}{32} = 1.0$$

$$= \frac{2m}{32} + \frac{4m}{32} + \frac{8m}{32} + \frac{16m}{32} + \frac{m}{32} = 1.0$$

$$2m + 4m + 8m + 16m + m = 1.0 \times 32$$

$$= \frac{31m}{32} = 1$$

$$m = \frac{32}{31} \neq$$

Exercise 3.5

$$\text{PDF is } f(n) = \frac{\binom{4}{n}}{16} \quad n = 0, 1, 2, 3, 4$$

n	PDF	CDF
0	$\frac{1}{16} = 0.0625$	0.0625
1	$\frac{4}{16} = 0.25$	$0.0625 + 0.25 = 0.3125$
2	$\frac{6}{16} = 0.375$	$0.3125 + 0.375 = 0.6875$
3	$\frac{4}{16} = 0.25$	$0.6875 + 0.25 + 0.0625 = 0.9375$
4	$\frac{1}{16} = 0.0625$	$0.9375 + 0.0625 = 1$

$$f(n) = \begin{cases} 0, & n < 0 \\ 0.0625, & 0 \leq n < 1 \\ 0.3125, & 1 \leq n < 2 \\ 0.6875, & 2 \leq n < 3 \\ 0.9375, & 3 \leq n < 4 \\ 1, & n \geq 4 \end{cases}$$

$$f(n) = n < 0 = 0$$

$$f(n) = 0 \leq n < 1 = P(n=0) = \frac{1}{16}$$

$$f(n) = 1 \leq n < 2 = P(n=0) + P(n=1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$$

$$f(n) = 2 \leq n < 3 = P(n=0) + P(n=1) + P(n=2) = \frac{1}{16} + \frac{4}{16} + \frac{6}{16} = \frac{11}{16}$$

$$f(n) = 3 \leq n < 4 = P(n=0) + P(n=1) + P(n=2) + P(n=3) = \frac{1}{16} + \frac{4}{16} + \frac{6}{16} + \frac{4}{16} = \frac{15}{16}$$

$$f(n) = n \geq 4 = 1$$

3 EXERCISE 3-6

n	0	1	2	3	4	5	6
$P(n=n)$	0.029	0.049	0.078	0.155	0.212	0.262	0.215
$n \cdot P_n$	0	0.049	0.156	0.465	0.848	1.31	1.29

$$a) \sum (n \cdot P_n) = 0 + 0.049 + 0.156 + 0.465 + 0.848 + 1.31 + 1.29$$

$$E(n) = 4.118$$

$$b) \sum (n^2) = n^2 \cdot (P_n = n)$$

$$= (0^2 \times 0.029) + (1^2 \times 0.049) + (2^2 \times 0.078) + (3^2 \times 0.155) + (4^2 \times 0.212) + (5^2 \times 0.262) + (6^2 \times 0.215) = 19.438$$

$$\text{var} = 19.438 - (4.118)^2 = 2.4801$$

$$\text{stdev} = \sqrt{2.4801} = 1.5748$$

Exercise 3-12 (Poisson) $P(n) = \frac{\lambda^n e^{-\lambda}}{n!}$

200 - typographical errors

500 - page manuscript

- three given errors = $(n = 0, 1, 2, 3)$

$$\lambda = 200$$

$$1 \text{ page} = \frac{\lambda}{500} = \frac{200}{500} = \frac{2}{5} = \lambda = 0.4$$

$$P(n) = \frac{\lambda^n e^{-\lambda}}{n!}$$

$$= \frac{0.4^3 \cdot e^{-0.4}}{3!} = 0.0042$$

Exercise 3.10

- There are 48 jackets for sale, 12 are black / non-black = 36
- Find probability

a) Exactly three will be black

$$P(X=3) = \frac{\binom{K}{n} \binom{N-K}{n-n}}{\binom{N}{n}}$$

$$P(n=3) = \frac{\binom{12}{3} \cdot \binom{36}{3}}{\binom{48}{6}} = \frac{12!}{3! (12-3)!} = \frac{12!}{3! \cdot 9!}$$

$$\binom{48}{6}$$

$$= \frac{12!}{2177280} = 220$$

$$\binom{36}{3} = \frac{36!}{3! (36-3)!} = \frac{36!}{3! \cdot 33!} = 7140$$

$$\binom{48}{6} = \frac{48!}{6! (48-6)!} = 12271512$$

b) more than two will be black

$$P(n=3, 4, 5, 6)$$

$$P(n > 2) = P(n=3) + P(n=4) + P(n=5) + P(n=6)$$

$$P(n=3) = 0.1280$$

$$P(n=4) = \frac{\binom{12}{4} \binom{36}{2}}{\binom{48}{6}} = \frac{12!}{4! (12-4)!} = 495 \quad \frac{495 \cdot 630}{12271512} = 0.0254\%$$

$$\frac{36!}{2! (36-2)!} = 630$$

$$\frac{48!}{6! (48-6)!} = 12271512$$

$$P(n=5) = \frac{\binom{12}{5} \binom{36}{1}}{\binom{48}{6}} = \frac{12!}{5!(12-5)!} = 792 = \frac{48!}{6!(48-6)!} = 12271512$$

$$= \frac{36!}{1!(36-1)!} = 36$$

$$\frac{792 \cdot 36}{12271512} = 0.0233 \quad \#$$

$$P(n=6) = \frac{\binom{12}{6} \binom{36}{0}}{\binom{48}{6}} = \frac{12!}{6!(12-6)!} = 924$$

$$= \frac{36!}{0!(36-0)!} = 1$$

$$= \frac{924 \cdot 1}{12271512} = 0.075 \quad \#$$

$$P(n \geq 2) = 0.0254 + 0.0233 + 0.075 + 0.1280$$

$$= 0.2517$$

c) at least one will be black

$$P(n=0, 1, 2, 3, 4, 5, 6)$$

$$P(n=0) = \frac{\binom{12}{0} \binom{36}{6}}{\binom{48}{6}} = \frac{12!}{0!(12-0)!} = 1$$

$$= \frac{36!}{6!(36-6)!} = 1947792$$

$$\frac{1 \cdot 1947792}{12271512} = 0.1587$$

$$1 - P(n=0) = 1 - 0.1587 = 0.8413 \quad \#$$

Exercise 8-8 (85% percent of dishwashers manufactured by a large company)

$$P(X=k) = \binom{n}{k} p^n q^k$$

$$a) P(X=0) = \binom{5}{0} p^5 q^0 = (1)(0.85)^5 (1) = 0.4437$$

b) Three or more will need repairs within two years.

$$\begin{aligned} \rightarrow P(X=3) &= \binom{6}{3} \cdot (0.85)^3 \cdot (0.15)^3 \\ &= 120 \cdot (0.85)^3 \cdot (0.15)^3 = 0.0139 \end{aligned}$$

$$\begin{aligned} \rightarrow P(X=4) &= \binom{5}{4} \cdot (0.85)^4 \cdot (0.15)^1 \\ &= 5 \cdot (0.85)^4 \cdot (0.15)^1 = 0.0026 \end{aligned}$$

$$\begin{aligned} \rightarrow P(X=5) &= \binom{5}{5} \cdot (0.85)^5 \cdot (0.15)^0 \\ &= 1 \cdot (0.15)^5 = 0.0001 \end{aligned}$$

$$P(X \geq 3) = 0.0139 + 0.0026 + 0.0001 = 0.0266$$

c) At most two will need repairs within two years

$$\rightarrow P(X=0) = 0.4437$$

$$\begin{aligned} \rightarrow P(X=1) &= \binom{5}{1} \cdot (0.85)^4 \cdot (0.15)^1 \\ &= 5 \cdot (0.85)^4 \cdot (0.15)^1 = 0.3915 \end{aligned}$$

$$\begin{aligned} \rightarrow P(X=2) &= \binom{5}{2} \cdot (0.85)^3 \cdot (0.15)^2 \\ &= 10 \cdot (0.85)^3 \cdot (0.15)^2 = 0.1382 \end{aligned}$$

$$P(X \leq 2) = 0.4437 + 0.3915 + 0.1382 = 0.9734$$

d) From one to four will need repairs

$$P(1 \leq n \leq 4) = 0.3915 + 0.1382 + 0.0382 + 0.0139 + 0.0026 = 0.5844$$